

Guide 32 - Welder and Fabrication Technician

Language: English

Version: 2.0 controlled revision

Revision date: August 11, 2026

Purpose: Practical career-entry and advancement guide for welders, fabrication technicians, welding helpers, fitter-welders, and closely related metal-fabrication roles.

This guide is educational. It does not guarantee employment, wages, admission, funding, reimbursement, apprenticeship placement, certification, licensing, promotion, or legal authority to perform welding or fabrication work. Requirements vary by jurisdiction, employer, process, material, position, code, contract, and worksite. Verify current rules and competent supervision before paying for training or performing hazardous work.

1. What this work is and who it may fit

Welders and fabrication technicians join, cut, fit, prepare, inspect, and repair metal components using approved processes, drawings, procedures, tools, and safety controls. Depending on the job, work may involve structural steel, sheet metal, piping, pressure equipment, vehicles, manufacturing assemblies, repair work, industrial maintenance, construction, shipbuilding, energy facilities, or custom fabrication.

This field can fit people who enjoy practical problem-solving, measurement, tools, drawings, precision work, visible results, and learning through supervised practice. It also requires disciplined safety judgment. Welding and cutting can expose workers to fire, explosion, electric shock, ultraviolet and infrared radiation, hot metal, fumes, gases, compressed cylinders, noise, confined spaces, falls, sharp edges, heavy materials, and hazardous coatings or base metals.

A job title alone does not authorize every welding task. A production welder, structural welder, pipe welder, fitter-welder, welding helper, robotic-welding operator, maintenance welder, and fabrication technician may have different qualification, code, employer, owner, contract, or jurisdiction requirements.

2. What workers actually do

Typical duties may include:

- reading engineering drawings, weld symbols, dimensions, specifications, work orders, and approved welding procedure information;
- measuring, marking, fitting, clamping, tack-welding, and preparing components for assembly;
- selecting approved consumables, equipment settings, shielding gases, and joint-preparation methods under the applicable procedure and supervision;
- using authorized welding, cutting, grinding, drilling, bending, and material-handling equipment;

- checking fit-up, alignment, dimensions, distortion, surface condition, and visible weld quality;
- cleaning and preparing metal surfaces while controlling dust, fumes, coatings, and ignition hazards;
- maintaining traceability, inspection, repair, production, and quality records when required;
- coordinating with fitters, inspectors, engineers, supervisors, machinists, maintenance teams, quality staff, and other trades;
- stopping when the work requires a qualification, procedure, permit, competent supervision, respiratory-protection controls, confined-space program, engineering decision, or code authority that is not in place.

Title distinctions matter

Welder usually emphasizes manual or semiautomatic joining and cutting processes.

Fabrication technician is an employer-dependent title that may combine layout, fitting, cutting, forming, drilling, grinding, welding, assembly, and inspection.

Fitter-welder usually combines component fit-up with welding.

Welding operator may refer to mechanized, automated, or robotic equipment rather than the same scope as a manual welder.

Do not assume one title's wage, certification, or authority applies to another.

3. Safety, hot-work, fume, and task boundaries

Welding and cutting are hot-work activities with potentially severe consequences. OSHA 29 CFR 1910 Subpart Q addresses welding, cutting, and brazing in U.S. general industry. OSHA 1910.252 addresses fire prevention, protection, ventilation, confined spaces, and related controls. OSHA 1910.253 addresses oxygen-fuel gas welding and cutting, including hazards from unsafe fuel-gas and oxygen mixtures. OSHA 1910.254 requires workers designated to operate arc-welding equipment to be properly instructed and qualified for that equipment.

Fire and explosion

Sparks, slag, hot metal, arcs, flames, and heated surfaces can ignite combustible material or flammable atmospheres. Never treat a visually clear area as automatically safe for hot work. Follow the site's hot-work permit, fire-watch, isolation, gas-testing, housekeeping, and fire-protection controls where required.

Fumes, gases, coatings, and base metals

Welding fume composition depends on process, consumable, base metal, coatings, surface contamination, shielding gases, and work environment. Do not grind, heat, cut, or weld unknown or coated material until the hazard has been assessed. Ventilation, local exhaust, respiratory protection, medical evaluation, fit testing, exposure monitoring, or other controls may be required.

Compressed-gas cylinders

Cylinders contain stored energy and require correct storage, handling, separation, regulators, hoses, valves, and protection. Oxygen must not be treated as compressed air, and fuel-gas/oxygen mixtures can create explosion hazards.

Electric shock and radiation

Arc-welding equipment can present electrical hazards, especially in wet, confined, conductive, or damaged-equipment conditions. Welding arcs also emit intense optical radiation. Use the correct PPE and equipment condition checks and follow the manufacturer's and employer's procedures.

Confined spaces and difficult locations

Confined-space welding may require atmospheric testing, ventilation, attendants, permits, communication, rescue planning, equipment controls, and other protections. Never enter or weld in a tank, vessel, pit, enclosed structure, or similar space based only on personal judgment.

Stop and escalate when

- the welding procedure, material, filler metal, position, joint, preheat, interpass limits, or acceptance criteria are unclear;
- the job requires a qualification or certification you do not hold;
- coatings, residues, process contents, or prior service are unknown;
- hot-work permits, gas testing, ventilation, fire watch, respiratory controls, or required PPE are missing;
- cylinders, leads, torches, regulators, hoses, grounding/work-return arrangements, or electrical equipment appear damaged or improper;
- the work involves confined spaces, elevated work, energized systems, pressure equipment, critical structures, hazardous metals, or specialized code work outside your verified competence;
- a customer or supervisor asks you to bypass an inspection, procedure, safety device, engineering requirement, or code requirement.

4. Entry routes and transferable skills

The U.S. Bureau of Labor Statistics identifies a high school diploma or equivalent as typical entry education for welders, cutters, solderers, and brazers, together with technical and on-the-job training. BLS identifies vocational-technical institutes, community colleges, private welding schools, military training, and employer-based apprenticeship as possible pathways.

Common routes include:

- paid apprenticeship or structured employer training;
- welding helper, fabrication helper, fitter helper, or trainee roles with supervised progression;
- public technical schools and community colleges;
- carefully evaluated private welding schools;

- military welding or metalworking training translated honestly into civilian requirements;
- manufacturing, machining, maintenance, construction, automotive, sheet-metal, or metal-fabrication experience;
- public vocational programs such as SENA welding-related training in Colombia.

Useful transferable skills include:

- measurement, fractions, decimals, geometry, and unit conversion;
- reading drawings and weld symbols;
- hand- and power-tool discipline;
- visual attention and dimensional accuracy;
- mechanical reasoning and troubleshooting;
- material identification and traceability;
- documentation and quality-control habits;
- teamwork and communication;
- willingness to stop when the procedure, safety condition, or scope is uncertain.

Do not claim Certified Welder, Red Seal, code-qualified, pressure-welder, structural-welder, pipe-welder, inspector, or other status you do not actually hold.

5. United States labor-market evidence

Official national statistics

BLS reports for Welders, Cutters, Solderers, and Brazers:

- **May 2024 median pay: \$51,000 per year / \$24.52 per hour;**
- **projected employment growth, 2024-2034: 2%;**
- **average projected openings: about 45,600 per year** over the decade, with many openings expected from replacement needs.

These are national statistics for the combined BLS occupational group. They are not fabrication-technician-only, apprentice, local-market, union-scale, project-specific, or guaranteed wage figures.

Current non-government market estimate

Salary.com's welder salary page, with a snapshot dated **July 1, 2026**, reported a U.S. average of approximately 59,272 *per year* (about **\$28 per hour**) and a displayed 25th-75th percentile range of approximately 52,614 – **67,828 per year**.

This is a **non-government market estimate** based on Salary.com's methodology. It is not definition-equivalent to BLS data and must not be presented as an official federal wage. Use it only as a current market cross-check.

When comparing offers, ask about shift differentials, overtime, travel, per diem, tools, PPE, respirator requirements, testing fees, certification continuity, union dues, production incentives, health benefits, and paid training.

6. United States training, funding, and free-first pathways

Before taking private-school debt, compare lower-cost and paid pathways.

Registered Apprenticeship and employer training

Apprenticeship.gov provides a federal Apprenticeship Job Finder that can be used to search open apprenticeship opportunities. Treat it as a locator and verification starting point. Do not assume every advertisement using the word "apprentice" is a Registered Apprenticeship Program.

For any program, verify the sponsor, registered occupation when applicable, paid status, starting wage, wage progression, related instruction, work hours, testing, tools/PPE, cancellation terms, and whether completion leads to a recognized credential.

WIOA and American Job Centers

WIOA-supported services can include occupational skills training, on-the-job training, workplace instruction, skills upgrading, retraining, and customized training when local eligibility rules are met. American Job Centers can help users investigate local services.

Ask in writing:

- whether the welding program is on the applicable Eligible Training Provider List when required;
- whether you are eligible before enrollment;
- whether approval must occur before costs are incurred;
- whether tuition, testing, books, tools, PPE, transportation, childcare, or supportive services are covered;
- whether funds remain available for the current program period.

Never assume WIOA will pay simply because a course is occupational.

Community college, technical school, and scholarships

Public community and technical colleges may offer lower-cost welding certificates, diplomas, or associate-level options. Compare total cost, lab hours, process coverage, instructor access, safety controls, employer connections, placement claims, and whether the program aligns with local employer and code requirements.

Scholarships from schools, employers, foundations, unions, trade groups, or welding organizations may exist, but award rules and deadlines change. Verify current eligibility and never pay a fee merely for access to a scholarship application.

Employer support

Ask employers about paid trainee roles, incumbent-worker training, tuition reimbursement, certification-test reimbursement, paid study time, tool/PPE support, and shift arrangements. Obtain written terms before enrolling or assuming reimbursement.

7. AWS Certified Welder and credential boundaries

The American Welding Society describes its **Certified Welder** credential as a performance-based industry certification administered through AWS Accredited Testing Facilities. AWS states that the program does not impose one universal prerequisite course or minimum experience requirement; the candidate must successfully complete the applicable performance qualification test.

Testing is procedure-, process-, material-, position-, and code/standard-dependent. Certification continuity and maintenance requirements apply.

Important boundaries:

- AWS certification is an **industry credential**, not a universal government license to weld anything anywhere;
- one successful performance qualification does not automatically qualify a welder for every process, position, thickness, diameter, material, filler metal, joint, code, or employer;
- employers, project owners, contracts, codes, and jurisdictions may require additional testing or qualifications;
- a school completion certificate is not automatically an AWS Certified Welder credential;
- verify the current AWS rules, testing facility, procedure, fees, continuity requirements, and employer recognition before paying.

8. Canada pathway

Red Seal - Welder

Red Seal identifies **Welder** as a Red Seal trade associated with **NOC 72106**. The Red Seal Occupational Standard provides a national trade standard and supports Red Seal examination content.

Apprenticeship registration, compulsory or voluntary trade status, certification, exams, and legal work requirements remain provincial or territorial. Verify the authority where you will train and work rather than assuming one national rule applies everywhere.

Government of Canada Job Bank wages

The current national Job Bank wage page for welders and related machine operators, NOC 72106, shows approximately:

- **C\$22.00/hour low;**
- **C\$30.00/hour median;**

- **C\$47.00/hour high.**

The page identifies a November 19, 2025 update. These are national reference values, not guaranteed offers. Regional, union, industrial, remote-site, shift, and project wages can differ substantially.

Apprentice support

Eligible apprentices in designated Red Seal trades may currently access the **Canada Apprentice Loan**, which offers up to **C\$4,000 in interest-free loans per period of technical training**, subject to current eligibility rules. Quebec has alternate support arrangements. Federal pages also direct apprentices to possible Employment Insurance support during eligible technical training and other skilled-trades supports.

Do not rely on outdated references to the former Apprenticeship Incentive Grant or Apprenticeship Completion Grant; those programs closed and should not be presented as currently available cash grants.

9. Colombia pathway

SENA training

SENA's Betowa platform lists welding-related training when cohorts are available, including courses focused on processes such as SMAW on carbon steels. Availability, city, schedule, modality, admission conditions, and capacity change by cohort.

Use the current Betowa catalog rather than assuming a specific course remains open. Short complementary courses can be useful skill-building steps but should not be represented as equivalent to a multi-year apprenticeship or as automatic authorization for specialized welding work.

SENA Agencia Publica de Empleo

SENA's Agencia Publica de Empleo (APE) is a free public employment service that lists welding-related vacancies when employers post them. Titles visible in 2026 include general welder, TIG welder, welding helper, fitter-welder or pailero-soldador, pipe welder, and specialized welding roles.

Use APE to compare actual employer requirements, locations, contract types, experience expectations, and salary bands when supplied. A single vacancy is not a national wage benchmark and may expire quickly.

Colombia work-boundary rule

Do not assume an AWS or foreign credential automatically replaces Colombian employer, project, safety, technical, contractual, or regulatory requirements. For structural, pressure, pipeline, industrial, construction, or other specialized work, verify the applicable technical standards, client specifications, employer qualification tests, safety rules, and any required competent or professional oversight.

10. Latin America beyond Colombia

Welding titles, public training systems, labor rules, technical standards, project specifications, safety regimes, and qualification requirements differ across Latin America.

For any country:

1. identify the labor, occupational-safety, technical-education, and public-employment authorities;
2. check public vocational institutes before paying a private provider;
3. verify whether welding qualifications are employer-, code-, project-, sector-, or regulator-specific;
4. confirm recognition of foreign credentials and experience;
5. verify local wage data separately from U.S. or Canadian data;
6. compare apprenticeship, trainee, dual-education, or employer-sponsored routes where available;
7. document which rules apply to the actual worksite and process.

Do not transplant U.S. OSHA rules, Canadian Red Seal requirements, Colombian SENA programs, or AWS credential assumptions into another jurisdiction without verification.

11. Skills to build deliberately

Technical foundations

Develop competence in:

- measurement, fractions, decimals, geometry, angles, and unit conversion;
- reading fabrication drawings, weld symbols, dimensions, tolerances, and basic bills of material;
- joint design, fit-up, tack sequence, distortion awareness, and alignment;
- material identification and surface preparation;
- approved use of SMAW, GMAW/MIG, FCAW, GTAW/TIG, oxy-fuel, plasma, or other processes only within actual training and work scope;
- cutting, grinding, drilling, forming, and finishing under applicable controls;
- visual inspection basics without claiming authority beyond training;
- consumable storage and traceability where required;
- safe material handling and shop organization.

Professional skills

Build habits in:

- punctuality and production reliability;
- clear communication with supervisors and inspectors;
- accurate work-order and quality documentation;
- asking before deviating from a drawing or procedure;

- reporting defects instead of hiding them;
- keeping qualification and continuity records current;
- learning from rework, inspection findings, and supervised practice.

12. A safe beginner learning sequence

A practical sequence is:

1. **Safety first:** PPE, hot-work controls, fume hazards, cylinder handling, electrical awareness, fire prevention, housekeeping, and emergency procedures.
2. **Measurement and drawings:** fractions, decimals, units, angles, dimensions, weld symbols, and basic fabrication drawings.
3. **Shop fundamentals:** material identification, layout, marking, cutting, grinding, drilling, clamping, and fit-up under supervision.
4. **Process fundamentals:** learn one welding process in a controlled training environment before adding more processes.
5. **Position and joint progression:** practice from simpler controlled coupons toward the positions and joints required by the target job or qualification.
6. **Inspection habits:** learn to identify obvious discontinuities, dimensional problems, poor fit-up, and procedural deviations without claiming inspector authority.
7. **Procedure discipline:** understand that a welding procedure specification or equivalent approved instruction governs essential job variables where required.
8. **Employment alignment:** match practice to real local job requirements rather than collecting unrelated certificates.
9. **Qualification preparation:** test only when the target credential, process, position, code, and employer need are clear.
10. **Continuous improvement:** keep a log of supervised practice, test results, rework causes, and next learning goals.

Never use an unsupervised home hot-work project as the default first step.

13. Portfolio and evidence without unsafe demonstrations

A welding portfolio should demonstrate judgment as well as craft. Useful evidence may include:

- photographs of completed training coupons or projects where photography is allowed;
- drawings or fabrication plans you are permitted to share;
- dimensional-inspection worksheets;
- training records and verified course completions;
- qualification records you are authorized to present;
- documented process improvement or reduced rework where evidence exists;
- before/after fabrication examples that contain no employer-confidential information;
- a skills matrix clearly separating trained, supervised, qualified, and not-yet-practiced tasks.

Do not publish proprietary drawings, customer information, restricted facility images, controlled technical data, or safety-sensitive information.

14. Ethical use of AI, privacy, and cybersecurity

AI can help explain terminology, create study questions, summarize public reference material, compare career pathways, organize a study plan, or improve a resume. It must not replace the controlled technical information required for actual welding work.

Appropriate uses

- explain the difference between common welding processes at a conceptual level;
- quiz you on weld symbols or safety vocabulary;
- help convert units while you independently verify critical calculations;
- organize public job postings into a skills matrix;
- help draft questions for a school, apprenticeship sponsor, employer, or testing facility;
- improve plain-language documentation after removing confidential information.

Unsafe or inappropriate uses

Do not rely on AI to:

- invent or alter a WPS, PQR, WPQ, engineering drawing, code requirement, inspection acceptance criterion, or manufacturer instruction;
- choose parameters for a critical weld without the approved procedure and qualified supervision;
- decide that a confined space, coated material, cylinder setup, pressure component, or hot-work area is safe;
- fabricate certification, continuity, inspection, test, employment, or training records;
- upload proprietary drawings, customer data, restricted facility information, export-controlled information, or confidential procedures to an unapproved AI service.

Treat AI output as unverified until checked against authoritative sources and workplace controls.

15. Evaluating schools and training providers

Before paying, ask for written answers to these questions:

- What exact credential or completion document is awarded?
- Which welding processes, positions, materials, thickness ranges, and joint types are actually taught?
- How many supervised lab hours are included?
- What is the student-to-booth or student-to-instructor ratio?
- Are consumables, PPE, testing, books, tools, and retests included in the advertised price?
- Does the program prepare for a specific industry qualification, and who administers the test?
- Is the testing facility currently recognized by the credentialing body where claimed?

- What employers hire graduates, and how were placement figures calculated?
- Are wage claims based on verified graduates, national averages, or marketing estimates?
- What refund, withdrawal, retest, transfer, and complaint policies apply?
- Can public funding, employer reimbursement, apprenticeship, or lower-cost technical-college options reduce the cost?

Avoid providers that guarantee jobs, wages, certification, or "universal" qualification.

16. Job-search strategy

Search by both occupation and process. Useful terms can include:

- welder;
- welding helper;
- fabrication technician;
- metal fabricator;
- fitter-welder;
- structural welder;
- pipe welder;
- TIG welder;
- MIG welder;
- FCAW welder;
- maintenance welder;
- welding operator;
- production welder.

Read the full posting. Record required process, material, position, code, certification, test, shift, travel, lifting, respirator, site-access, and experience requirements.

Apply honestly. If a posting asks for a qualification you do not hold, state related training accurately rather than implying certification. When a practical test is required, ask which process, position, material, thickness/diameter, code or procedure, and PPE/testing conditions will apply.

17. Interview and practical-test preparation

Be prepared to discuss:

- how you verify the correct drawing and procedure before starting;
- how you handle unclear dimensions or weld symbols;
- how you prevent fire and control hot-work hazards;
- how you respond to unknown coatings or fume concerns;
- how you protect cylinders and inspect equipment;
- how you check fit-up and dimensions before welding;
- what you do when a weld does not meet the required condition;

- which processes, materials, positions, and thicknesses you have actually practiced or qualified on;
- a time you stopped work or escalated a safety or quality concern.

For a practical test, follow the test instructions exactly. Do not improvise essential variables to "show initiative" when a procedure or examiner controls them.

18. Career progression

Possible pathways include:

- welding helper or trainee;
- production or fabrication welder;
- fitter-welder;
- structural or pipe welding after appropriate qualification;
- maintenance welding;
- robotic/automated welding operator;
- fabrication lead or working supervisor;
- welding quality or inspection support with additional training;
- welding inspector or welding supervisor only after meeting the relevant credential and experience requirements;
- welding engineering or engineering-technology pathways with additional formal education where required;
- training, apprenticeship instruction, estimating, planning, or shop management.

Progression should be based on documented competence, safe performance, and real qualification rather than title inflation.

19. A 90-day action plan

Days 1-30 - verify and choose

- Identify the actual welding/fabrication jobs available in your location.
- Record the recurring processes, materials, positions, shifts, and qualifications employers request.
- Compare public/community technical programs, employer trainee roles, apprenticeship options, and private schools.
- Check funding before paying: employer support, WIOA/American Job Center routes in the U.S., provincial/territorial apprentice support in Canada, SENA in Colombia, and public vocational options elsewhere in Latin America.
- Complete foundational safety and measurement study.

Days 31-60 - train deliberately

- Begin supervised shop and process training.
- Build measurement, drawing, fit-up, and equipment-inspection habits.
- Keep a practice log with process, material, position, joint, settings as authorized by the instructor/procedure, observed defects, and corrective coaching.

- Update your resume using only skills you can support.
- Investigate the exact qualification test that matches target employers.

Days 61-90 - demonstrate readiness

- Complete supervised projects or coupons appropriate to your level.
- Practice explaining safety, procedure discipline, and quality decisions.
- Apply to roles whose requirements match your verified skills.
- Schedule a qualification test only when the credential and employer need are clear.
- Continue building toward the next process, position, or responsibility instead of collecting unrelated certificates.

20. Decision checklist before spending money

Before paying for a welding program, certification test, tool package, or private course, verify:

- the target job and local demand;
- the exact process, material, position, code, or qualification employers expect;
- whether the provider teaches that requirement;
- total tuition and all additional fees;
- whether PPE, tools, consumables, books, testing, and retests are included;
- public funding, scholarships, apprenticeship, employer support, or reimbursement;
- refund and withdrawal terms;
- current testing-facility status where a credential is claimed;
- whether a cheaper public or employer-supported route reaches the same goal;
- whether the credential actually matters to target employers.

Delay or decline enrollment when the provider will not document its claims, guarantees employment or wages, hides total cost, pressures immediate financing, cannot explain the testing standard, or implies one certificate qualifies you for all welding work.

21. Current source and verification list

Use current official pages whenever possible:

United States

- U.S. Bureau of Labor Statistics, Welders, Cutters, Solderers, and Brazers:
<https://www.bls.gov/ooh/production/welders-cutters-solderers-and-brazers.htm>
- OSHA Welding, Cutting and Brazing, 29 CFR 1910 Subpart Q:
<https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910SubpartQ>
- OSHA 1910.252 general requirements:
<https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.252>
- OSHA 1910.253 oxygen-fuel gas welding and cutting:
<https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.253>
- OSHA 1910.254 arc welding and cutting:
<https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.254/>

- Apprenticeship.gov Job Finder: <https://www.apprenticeship.gov/apprenticeship-job-finder>
- U.S. Department of Labor WIOA programs: <https://www.dol.gov/agencies/eta/wioa/programs?lang=en>
- American Welding Society Certified Welder: <https://www.aws.org/Certification-and-Education/Professional-Certification/Certified-Welder-Program/>

Non-government U.S. market cross-check

- Salary.com welder salary estimate, snapshot dated July 1, 2026: <https://www.salary.com/research/salary/opening/welder-salary>

Canada

- Red Seal Welder: <https://red-seal.ca/eng/trades/welder.shtml>
- Red Seal Welder occupational overview: <https://red-seal.ca/eng/trades/weld/overview.shtml>
- Government of Canada Job Bank wage report, NOC 72106: <https://www.jobbank.gc.ca/wagereport/occupation/23308>
- Canada skilled-trades apprentice support: <https://www.canada.ca/en/services/jobs/training/support-skilled-trades-apprentices/become-apprentice.html>
- Canada Apprentice Loan: <https://www.canada.ca/en/services/jobs/training/support-skilled-trades-apprentices/loan.html>
- Skilled-trades funding opportunities: <https://www.canada.ca/en/services/jobs/training/support-skilled-trades-apprentices/funding-opportunities.html>

Colombia

- SENA Betowa: <https://betowa.sena.edu.co/>
- SENA Agencia Publica de Empleo: <https://agenciapublicadeempleo.sena.edu.co/>

22. Final reminder

A strong welding career is built through supervised practice, procedure discipline, safety, accurate representation of qualifications, and repeated verification of current requirements. Start with the lowest-cost credible pathway that matches real employer demand. Treat every credential as scope-specific. Keep safety, quality, and documentation ahead of speed.

Created and directed by Alberto "Al" Leiva. ChatGPT assisted with research organization, editing, and document preparation under the author's direction. The author remains responsible for editorial and publication decisions.

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